# Technical Document #312: Deep Learning - Comprehensive Overview

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Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Convolutional neural networks (CNNs) are designed to process grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow. They were introduced in the ResNet architecture.

Dropout is a regularization technique where randomly selected neurons are ignored during training. This prevents co-adaptation and improves generalization.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

### Key Takeaways:

- Transfer learning is a technique where a model trained on one task is re-purposed on a second related task.
- Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions.
- Dropout is a regularization technique where randomly selected neurons are ignored during training.